

HelixQA Autonomous Sessions — HelixVPN nano-detail spec (Volume 8 · §11.4.169 type 6)

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Nano-detail expansion of [§5.6 of the Volume-8 overview](#). HelixQA (HelixDevelopment/helix_qa) is the **autonomous QA orchestrator** (§11.4.27 / §11.4.107 / §11.4.158): it drives the full MVP-DoD acceptance set end-to-end with **no human**, consumes the challenges banks ([challenges.md](#)), boots infra via containers ([integration.md](#)), drives the UI via panoptic and analyses frames via `vision_engine`, generates tickets for findings into the workable-items DB, respawns crashed sub-runs (§11.4.147), and emits the **real-time sync channel** (§11.4.116 — append-only JSONL event stream + atomically- rewritten status snapshot) that the conductor tails live. It is the §11.4.165 **independent-verification agent for the runtime layer** — structurally separate from the code author. This document fixes the harness, the written test banks per app/service/platform, the captured wire-evidence-per-check contract, the sync channel, the determinism, the gate, the paired §1.1 mutation, and skeletons. Spec-only; unproven assumptions marked UNVERIFIED. Siblings: [unit.md](#), [integration.md](#), [e2e.md](#), [full-automation.md](#), [challenges.md](#).

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1. Scope — what HelixQA orchestrates

HelixQA is not a single test type — it is the **autonomous QA session orchestrator** that drives every other type (CHAL, E2E, INT, SEC, REC) as an end-to-end run with no human, captures wire evidence per check, and reconciles the verdicts into the coverage ledger and the workable-items DB. Per §11.4.27 it must be used with **written test banks (suites) for every application, service, and platform** plus **fully-autonomous QA sessions** driving every registered bank with captured wire evidence per check.

HelixQA responsibility	Mechanism	Constitution
drive the DoD acceptance set end-to-end, no human	--suite mvp-dod over the challenges banks	§11.4.27 / §11.4.98
boot infra for the run	containers/pkg/boot (rootless)	§11.4.76 / §11.4.161
drive the UI flows	panoptic (web/desktop/mobile) → window-scoped MP4	§11.4.154 / §11.4.159
analyse frames for the REC verdict	vision_engine OCR + core-FSM cross-check	§11.4.163 / §11.4.165
emit a real-time sync channel	append-only JSONL events + atomic status snapshot	§11.4.116
respawn crashed sub-runs	durable agent registry, crash≠done	§11.4.147
generate tickets for findings	write into the §11.4.93/.95 workable-items DB	§11.4.148
be the independent verifier	structurally separate from the code author	§11.4.165

The **written test banks** are organised per surface:

Bank	Surface	Suites it holds
helix_qa/banks/control_plane.yaml	the Go services	enrollment, RLS, reconcile, revoke, events, API authz (drives INT + CHAL)
helix_qa/banks/data_plane.yaml	helix-core + edge	reach, default-deny, escalation, kill-switch, liveness (drives E2E + SEC)
helix_qa/banks/clients_<platform>.yaml	per-platform app	connect flow, status UI, exit picker — one per platform (Access/Connector/Console × iOS/Android/Linux/Windows/macOS/Web; HarmonyOS/Aurora Phase 3) the 8 ACs (challenges.md §8)

Bank	Surface	Suites it holds
helix_qa/banks/ mvp_dod.yaml	the release gate	

UNVERIFIED: the exact bank-file naming and the per-platform split granularity are the intended layout; the obligation (a written bank per app/service/platform, §11.4.27) is fixed, the file layout is a Phase-1 instantiation per §11.4.35.

2. Harness — the helix_qa orchestrator + test banks

helix_qa (HelixDevelopment/helix_qa, Go) is consumed as an owned submodule (§11.4.28, equal-codebase, consumed by reference / pinned SHA), invoked as one command ([overview §9](#)): `go run ./submodules/helix_qa/cmd/orchestrator --suite mvp-dod`.

```
graph TD
    Q["helix_qa orchestrator --suite mvp-dod"] --> R["agent registry  
(§11.4.147)<br/>dispatched|in-flight|crashed|respawned|complete"]
    Q --> I["containers boot infra (§11.4.76)"]
    Q --> RIG["rig/netns_up.sh (§11.4.3)"]
    Q --> CH["challenges engine: score each AC (§5.5)"]
    Q --> PA["panoptic: drive UI → window-scoped MP4 (§11.4.154)"]
    PA --> VE["vision_engine: OCR + core-FSM cross-check  
(§11.4.163)"]
    CH --> EV["events.jsonl (append-only) + status.json (atomic) –  
§11.4.116"]
    VE --> EV
    EV --> CD["conductor tails live (real-time sync)"]
    Q --> T["findings → workable-items DB tickets (§11.4.148)"]
    Q -->|crash detected| R
    R -->|respawn until complete| Q
```

- **Independent verifier (§11.4.165):** HelixQA is dispatched as a structurally-separate agent from the code author — it re-reads the captured evidence and produces its own verdict; a HelixQA PASS is the runtime-layer independent confirmation, never the author asserting their own work.
- **Crash-respawn (§11.4.147):** every sub-run (per-AC Challenge, per-platform UI drive) is a registry entry; a transient crash (rate-limit, socket-close, a device drop §11.4.144) flips it to crashed, keeps it OPEN, and respawns until complete. The session's done-condition is **not** satisfied while any entry is non-complete — a registry showing complete without landed evidence is a §11.4 PASS-bluff at the agent-lifecycle layer.
- **Self-meta-test (§11.4.32):** HelixQA plants a known-broken AC and asserts the session reports FAIL — proving the orchestrator itself does not bluff.

3. Fixtures — real system, autonomous drive, captured wire evidence

Fixture	Real or mock	Notes
the full stack (control plane + edge + client + apps)	real	HelixQA drives the deployed system, not a mock
infra (PG/Redis)	real , via containers	rootless on-demand boot
netns rig	real	the E2E substrate (e2e.md)
UI drive	real via panoptic	the app's own UI; §11.4.117 pixel oracle for non-introspectable surfaces
frame analysis	real vision_engine OCR	golden-good/bad self-validated (§11.4.107(10))
.env credentials	real token, out-of-band	§11.4.10 — the one permitted human step
mocks / manual steps	FORBIDDEN (§11.4.27 / §11.4.98)	HelixQA is the autonomous-session layer; a human step breaks it

HelixQA's autonomy is the §11.4.98 invariant at the orchestration layer: a session that requires a human to type, click, or trigger anything after startup is a PASS-bluff. A flow genuinely infeasible to drive autonomously is an honest operator_attended SKIP with a tracked migration item (§11.4.52), surfaced on the sync channel — never a faked PASS.

4. Evidence taxonomy — the §11.4.116 sync channel + per-check wire evidence

HelixQA's distinguishing evidence is the **real-time sync channel** plus a **captured wire artifact per check**. Per §11.4.116 the channel is two parts:

1. an **append-only JSONL event stream** (qa-results/helix_qa/events.jsonl), one event per line, never rewritten;
2. an **atomically-rewritten status snapshot** (qa-results/helix_qa/status.json, write-temp-then-rename) the conductor tails.

```
// qa-results/helix_qa/events.jsonl – append-only (§11.4.116)
{"ts":"2026-...", "ev":"session_start", "suite":"mvp-dod", "build":"<artifact-md5>"}
{"ts":"...", "ev":"challenge_start", "id":"HVPN-CHAL-AC2-authorized-reach"}
{"ts":"...", "ev":"evidence", "id":"HVPN-CHAL-AC2-...", "path":"qa-results/.../reach.pcap"}
{"ts":"...", "ev":"verdict", "id":"HVPN-CHAL-AC2-...", "result":"PASS", "evidence":"qa-results/.../reach.pcap"}
{"ts":"...", "ev":"agent", "id":"ui-drive-
```

```
ios","state":"crashed","reason":"device_dropped"} // §11.4.147
{"ts":"...", "ev":"agent", "id":"ui-drive-ios", "state":"respawned"}
```

The verdict-carries-evidence-path invariant (§11.4.116): a verdict event **must** carry the evidence path that backs it — a PASS event with no evidence path is a channel-layer PASS-bluff; a snapshot reporting PASS while the stream shows no evidence event for that AC is a **contradiction** → treated as FAIL; an AC with no challenge_start event cannot have a verdict. Verdicts use the closed vocabulary PASS / FAIL / SKIP / OPERATOR-BLOCKED (§11.4.45). The captured artifacts themselves are the §11.4.69 sink-side evidence per check ([challenges.md §4](#)).

The status snapshot:

```
// qa-results/helix_qa/status.json – atomically rewritten (write-
temp-then-rename)
{ "session":"mvp-dod","build":"<artifact-md5>","phase":"running",
  "counters":
{"pass":5,"fail":0,"skip":0,"operator_blocked":0,"pending":3},
  "last_verdict":{"id":"HVPN-CHAL-AC6-
revoke","result":"PASS","evidence":"qa-results/.../
revoke_timing.csv"},
  "open_agents":[{"id":"ui-drive-console","state":"in-
flight"}] } // §11.4.147 – not done while open
```

5. Determinism — N-iteration identical evidence

Per §11.4.50 a HelixQA session over the same artifact MD5 + same rig must produce an identical *verdict set* with structurally-identical per-check evidence across N=3 (normal) / N=10 (cycle-validation):

1. The orchestrator runs under the FA wrapper ([full-automation.md §5](#)); each AC's matched-evidence structural hash is compared across runs; a divergence is auto-FAIL.
 2. The sync-channel **event ordering** is deterministic per the risk-ordered schedule (§11.4.132 — security-floor ACs first); a reordered run with the same verdicts is acceptable, a *divergent verdict* is not.
 3. Crash-respawn does not break determinism: a respawned sub-run must reach the **same** verdict + same evidence shape as a first-attempt run; the registry proves the work was not lost, the evidence proves it was not corrupted (§11.4.147 honest boundary — durability, not correctness; correctness is the per-check evidence).
-

6. Acceptance gate — when HQA blocks a release

Gate	Bar	Layer
make qa (overview §9)	the mvp-dod session reports all 8 ACs PASS with per-check evidence	release sweep

Gate	Bar	Layer
CM-AUTONOMOUS-FRAMEWORK-SYNC-CHANNEL (§11.4.116)	the session emits the JSONL stream + atomic snapshot; every verdict carries an evidence path	pre-build + runtime
CM-HQA-NO-COMPLETE-WITHOUT-EVIDENCE (§11.4.147)	the done-condition is unsatisfied while any registry entry is non-complete; no complete without landed evidence	runtime
CM-HQA-SELF-META-TEST (§11.4.32)	the planted known-broken AC makes the session report FAIL	pre-build meta-test

HelixQA's session exit status **gates the release** — make qa is the §11.4.40 pre-tag full-suite-retest driver at N=10, and the §11.4.165 independent-verification gate (QA-D4: **independent re-read**, overview §10 — vision_engine re-reads the MP4 and cross-checks the core FSM, the harness's claim is never the verdict). A finding generates a workable-items ticket (§11.4.148); the feature's coverage-ledger cell stays out of AUTONOMOUS_VERIFIED until the session re-runs GREEN.

7. The paired §1.1 mutation (anti-bluff proof)

```
# §1.1 mutation for CM-AUTONOMOUS-FRAMEWORK-SYNC-CHANNEL (verdict-
carries-evidence)
- make the orchestrator emit a verdict event WITHOUT an `evidence`
path:
    {"ev":"verdict","id":"HVPN-CHAL-AC2-...","result":"PASS"}    //
no evidence path
- assert: the channel validator flags a PASS-with-no-evidence-path
as a contradiction
    → treats it as FAIL → gate FAILs (the bluff is caught)
- restore (evidence path required); assert PASS

# §1.1 mutation for CM-HQA-NO-COMPLETE-WITHOUT-EVIDENCE
(§11.4.147)
- mark a `crashed` registry entry as the session's done-condition
`complete` WITHOUT
    its sub-run landing evidence
- assert: the done-condition checker refuses to read satisfied →
session reports
    incomplete → gate FAILs (crash≠done enforced)
- restore (respawn-until-real-complete); assert the session
completes only with evidence

# §1.1 mutation for CM-HQA-SELF-META-TEST (§11.4.32 – the
orchestrator itself doesn't bluff)
- plant a known-broken AC (e.g. force default-deny to fail-open in
the test build)
- assert: the.mvp-dod session reports AC3 FAIL (the orchestrator
detects the break)
- a session that reports AC3 PASS on the broken build is itself
the bluff → meta-test FAILs
```

8. Test skeletons

8.1 The orchestrator invocation + sync-channel consumption

```
// submodules/helix_qa/cmd/orchestrator/main.go (invocation
//      contract; engine in submodule)
// go run ./submodules/helix_qa/cmd/orchestrator --suite mvp-dod
// 1. boot infra via containers (§11.4.76), bring up the rig
//      (§11.4.3)
// 2. for each AC in the mvp-dod bank, dispatch a sub-run
//      (registry entry §11.4.147)
// 3. score via the challenges engine (§5.5); drive UI via
//      panoptic; analyse via vision_engine
// 4. append every event to events.jsonl; rewrite status.json
//      atomically (§11.4.116)
// 5. on crash → flip registry to `crashed`, respawn until
//      `complete`
// 6. session PASS only when every AC verdict==PASS AND every
//      registry entry==complete-with-evidence

# the conductor's live tail of the sync channel (§11.4.116) –
#      never idle-blind (§11.4.94)
tail -F qa-results/helix_qa/events.jsonl | while read -r ev; do
    case "$(jq -r '.ev <<<"$ev"' in
        verdict)
        path=$(jq -r '.evidence // empty' <<<"$ev")
        [ -n "$path" ] || ab_fail "§11.4.116: verdict with no
            evidence path → contradiction (FAIL)"
        ;;
        agent) [ "$(jq -r '.state <<<"$ev"' = crashed ] &&
            note_open_workunit "$ev" ;; # §11.4.147
    esac
done
```

8.2 A control-plane bank suite entry (drives INT + CHAL)

```
# helix_qa/banks/control_plane.yaml – written test bank per
#      service (§11.4.27)
bank: helixvpn-control-plane
suites:
  - name: rls-isolation
    drives: [ "go test -tags integration -run
        TestRLSCrossTenantDenial ./helix-go/internal/store/..." ]
    evidence_class: rls_rowset # §11.4.69 –
        captured rowset
    capture: qa-results/int/rls_${RUN_ID}.json
  - name: revoke-latency
    drives: [ "rig/revoke.sh" ]
    evidence_class: counter_delta
    assert_p99_lt_seconds: 1.0 # SL03
    capture: qa-results/e2e/revoke_timing.csv
```

8.3 A per-platform client bank suite (drives UI + REC, §11.4.165 independent re-read)

```
# helix_qa/banks/clients_ios.yaml – written bank per platform
  (§11.4.27)
bank: helixvpn-client-ios
suites:
  - name: connect-flow
    drives: [ "panoptic drive --app access --platform ios --flow
              enroll-connect-reach" ]
    evidence_class: REC # §11.4.153/.159
    capture:
      kind: mp4
      window_scoped: true # §11.4.154 – app
                        window only
      filename_prefix: helixvpn--- # §11.4.155
      vision_verdict: vision_engine # §11.4.163
                        independent re-read (§11.4.165)
      expected_patterns:
        ["Connect","Connecting","Connected","shield:green"]
      cross_check: { core_fsm_at_green_shield: "Connected" } #
                  defeats B3
    on_uninstrumentable_ui: pixel_oracle # §11.4.117
                          fallback, never fake PASS
    on_no_device: skip_operator_attended # §11.4.3/.52
                  honest SKIP (real device required)
```

8a. The agent-lifecycle registry (§11.4.147 — crash ≠ done)

Every HelixQA sub-run (a per-AC Challenge, a per-platform UI drive, a device session) is a durable registry entry on the §11.4.116 sync substrate, so a crash never loses, forgets, or corrupts its work:

```
stateDiagram-v2
  [*] --> dispatched: orchestrator schedules the sub-run
  dispatched --> in_flight: agent claims it, drives the check
  in_flight --> complete: verdict + landed evidence (§11.4.69)
  in_flight --> crashed: rate-limit / socket-close / device drop
  (§11.4.144)
  crashed --> respawned: respawn until complete (autonomous
  §11.4.101)
  respawned --> in_flight: fresh agent, same scope, preserved
  state
    complete --> [*]
    note right of crashed
      the session done-condition is UNSATISFIED
      while ANY entry is non-complete (§11.4.87/.94/.147)
    end note
```


Lifecycle hazard	HelixQA response
transient API rate-limit kills a sub-run	flip to crashed, keep OPEN, respawn with the project's defined backoff (never invented, §11.4.6)
a tracked device drops mid-UI-drive	§11.4.144 availability-following: log an honest offline event, wait the defined reconnect budget, re-attach + resume, escalate on timeout
partial UI-drive state (half a recording)	preserve, §11.4.84 quiescence check, resume-or-clean-restart from a known-good base
a complete entry has no landed evidence	the done-condition refuses to read satisfied (a complete without evidence is a §11.4 PASS-bluff)

The honest boundary (§11.4.147): the registry + respawn guarantee the work is **not lost or corrupted**, NOT that it is **correct** — a respawned sub-run's output still crosses the per-check §11.4.69 evidence gate and the §11.4.40 retest; §11.4.147 is the durability layer beneath those.

8b. HelixQA as the independent verifier (§11.4.165) — the structural seam

HelixQA is dispatched as a **structurally-separate agent from the code author** (§11.4.70 / §11.4.165); §11.4.92 author-side self-evaluation PRECEDES but never satisfies it. The independence is load-bearing:

Verification axis	What HelixQA independently re-reads	The author claim it does NOT trust
reachability	the captured pcap (SYN-ACK present/absent)	"the reach test passed"
escalation	the tshark classification (H3 present, WG absent)	StatusReport.transport=="masque-h3" alone (B3)
UI flow	the window-scoped MP4 via vision_engine + core-FSM cross-check	"the shield went green" (a green shield while the FSM is Blocked FAILs)
no-log	the live-DB introspection report	"the schema is fine"

A HelixQA session that rubber-stamps the harness's claims without re-reading the artifacts is itself the bluff the §11.4.165 + §11.4.32 self-meta-test (§7) catches.

9. Open decisions surfaced for QA

#	Decision	Options	Recommendation
H-D1	Vision verdict authority (QA-D4)	trust the harness verdict vs independent vision_engine re-read	independent re-read (overview §10) — vision_engine re-reads the MP4 + cross-checks the core FSM (defeats B3); the harness claim is never the verdict (§11.4.165)
H-D2	crash-respawn backoff	fixed vs project-defined budget	project-defined backoff (§11.4.147) reusing the recovery path's existing budgets — never invented numbers (§11.4.6)
H-D3	per-platform bank granularity	one bank per platform vs one bank with platform matrix	one bank per platform for clear per-surface coverage (§11.4.27); a matrix view is a derived projection

UNVERIFIED: the helix_qa orchestrator CLI (`--suite`), the bank-YAML keys (`drives`, `evidence_class`, `vision_verdict`, `on_no_device`), and the `cmd/orchestrator` path are the **intended** shape per the overview skeleton; confirm against the pinned HelixDevelopment/helix_qa SHA (§11.4.74 — extend upstream if unsupported).

Sources verified

- [Volume-8 overview §2](#) (taxonomy row HQA), §5.6 (the orchestrator + the §11.4.116 sync-channel JSONL skeleton), §5.5 (banks it consumes), §7.2 (AC set it scores), §10 QA-D4 — read 2026-06-26.
- Sibling specs cross-referenced: [challenges.md](#), [e2e.md](#), [integration.md](#), [full-automation.md](#) — written 2026-06-26 in this same volume.
- Constitution: §11.4.27 (no-fakes / 100% type coverage / HelixQA banks + autonomous sessions), §11.4.116 (real-time sync channel / verdict-carries-evidence-path), §11.4.147 (crashed-agent respawn / crash≠done registry), §11.4.165 (independent verification agent), §11.4.32 (self-meta-test), §11.4.45 (closed verdict vocabulary), §11.4.69 (sink-side evidence per check), §11.4.98 (autonomous / no manual step), §11.4.117 (pixel-oracle fallback), §11.4.52/.3 (honest operator-attended SKIP), §11.4.153/.154/.155/.159/.163 (REC / window-scope / prefix / media-validation), §11.4.148 (ticket integrity), §11.4.93/.95 (workable-items DB), §11.4.50 (determinism), §11.4.40 (pre-tag retest), §11.4.144 (device-availability following), §1.1 (paired mutation) — from `CLAUDE.md` in-context.
- The helix_qa orchestrator CLI + bank-YAML schema are reproduced from the overview skeleton — **not independently fetched from HelixDevelopment/helix_qa** (UNVERIFIED; confirm at the pinned SHA).